■ Heat Pump Feasibility & Simulation Software: MVP Specification**

■ PART 1: DETAILED MVP SPECIFICATION**

■ 1. MVP Overview

1.1 Purpose

Develop a **web-based MVP** for heat pump feasibility studies and basic simulation, targeting **residential/commercial buildings** (greenhouse/CEA support deferred to Phase 2). The MVP will:

Enable **non-experts** (architects, contractors, homeowners) to **size and evaluate** heat pump systems.

Provide **quick feasibility checks** with thermal and economic outputs.

Serve as a **foundation** for future expansion (greenhouses, hybrid systems, advanced simulation).

1.2 Goals

1.3 Scope (Included in MVP)

■ User Inputs

Building dimensions (area, volume, orientation)

Envelope materials (walls, roof, floor, windows)

Location (auto-climate data)

Desired indoor temperature

Heat pump type (air-source, ground-source)

■ **Core Calculations**

Heat loss (BS EN 12831 or ASHRAE simplified)

Heat pump sizing (capacity, COP/SCOP)

Basic economic analysis (CAPEX, OPEX, simple payback)

■ **Outputs**

Heat load profile (kW)

Recommended heat pump size (kW)

Estimated COP/SCOP

Simple payback period (years)

Carbon savings (vs. gas boiler)

■ **Database**

Material properties (U-values, thermal conductivity)

Climate data (temperature, humidity)

Heat pump performance curves (generic models)

■ **UI/UX**

Web-based (responsive)

Guided input workflow

Visual results (charts, heat loss map)

1.4 Scope (Excluded from MVP)

■ **Greenhouse/CEA mode** (Phase 2)

■ **Hybrid systems** (solar + heat pump + storage) (Phase 2)

■ **Dynamic simulation** (hourly/transient) (Phase 3)

■ **Advanced economics** (NPV, IRR, incentives) (Phase 2)

■ **Collaboration features** (Phase 3)

■ **BIM/Revit integration** (Phase 4)

■ **IoT monitoring** (Phase 4)

■ 2. User Stories & Requirements

2.1 User Personas

2.2 Functional Requirements

FR-1: User Inputs

FR-2: Calculations & Simulation

FR-3: Economic Analysis

FR-4: Outputs & Visualization

FR-5: Database

FR-6: System & UX

■■ 3. System Architecture

3.1 High-Level Architecture

3.2 Component Diagram

3.3 Data Flow

User Inputs → Frontend validates → Sent to Backend (JSON).

Backend validates → Calls **Simulation Engine**.

Simulation Engine:

• Fetches **material/climate data** from Database.

• Runs **heat loss calculation**.

• Sizes **heat pump** and estimates **COP/SCOP**.

• Calculates **economics** (CAPEX, OPEX, payback).

Backend returns **results** (JSON) → Frontend renders.

Frontend displays **charts, tables, report**.

■■ 4. Data Model

4.1 Database Schema (PostgreSQL)

4.2 Sample Data (Materials)

4.3 Sample Data (Heat Pump Models)

■■ 5. Core Algorithms

5.1 Heat Loss Calculation (BS EN 12831 Simplified)

Formula:**

Where:

Q = Total heat loss (W)

U = U-value of surface (W/m²·K)

A = Area of surface (m²)

ΔT = Temperature difference (indoor - outdoor) (°C)

Q_ventilation = Heat loss from ventilation (W)

Q_infiltration = Heat loss from infiltration (W)

Steps:**

Calculate **transmission heat loss** for each surface (walls, roof, floor, windows).

Calculate **ventilation heat loss**:

```

Q_vent = 0.33 × n × V × ΔT

```

• n = Air changes per hour (default: 0.5 for residential)

• V = Volume of building (m³)

Sum all losses to get **total heat load (kW)**.

Example:**

Building: 100 m², 2.5m height → Volume = 250 m³

Walls: 200 m², U=0.3 W/m²·K

Roof: 100 m², U=0.2 W/m²·K

Windows: 20 m², U=2.8 W/m²·K

Outdoor temp: -5°C, Indoor temp: 20°C → ΔT = 25°C

Ventilation: 0.5 ACH

5.2 Heat Pump Sizing

Steps:**

Peak Heat Load = Q_total (from heat loss calculation).

Safety Factor = 1.15 (15% oversizing for cold climates).

Recommended Capacity = Q_total × Safety Factor.

Select Heat Pump = Smallest model with Capacity ≥ Recommended Capacity.

Example:**

Peak load = 4.4 kW

Recommended capacity = 4.4 × 1.15 = **5.06 kW**

Selected model: **Generic ASHP (Small, 5–10 kW)**

5.3 COP/SCOP Estimation

COP (Coefficient of Performance):**

Air-Source Heat Pump (ASHP):

```
COP = 0.05 × T_outdoor + 2.5 (Simplified linear model)
```

• Example: At 7°C → COP = 0.05×7 + 2.5 = **2.85** (real-world: ~3.5)

• At -5°C → COP = 0.05×(-5) + 2.5 = **2.25**

Ground-Source Heat Pump (GSHP):

• COP = **4.0** (constant, as ground temp is stable).

SCOP (Seasonal COP):**

Weighted average COP over the year (based on climate data).

Formula:

```
SCOP = Σ (COP_i × H_i) / Σ H_i
```

• COP_i = COP at temperature i

• H_i = Heating degree hours at temperature i

Example (Simplified):**

5.4 Economic Analysis

CAPEX:**

Heat Pump Cost = $/kW × Capacity (e.g., $1,500/kW)

Installation Cost = $/kW × Capacity (e.g., $1,000/kW)

Total CAPEX = Heat Pump Cost + Installation Cost

OPEX:**

Annual Energy Demand = Heat Load / COP × Runtime (kWh)

Annual Electricity Cost = Energy Demand × Electricity Price ($/kWh)

Savings (vs. Gas Boiler):**

Gas Boiler Efficiency = 0.9 (90%)

Annual Gas Demand = Heat Load / 0.9 × Runtime (kWh)

Annual Gas Cost = Gas Demand × Gas Price ($/kWh)

Annual Savings = Gas Cost - Electricity Cost

Simple Payback:**

Carbon Savings:**

■■ 6. API Specification

6.1 Endpoints

6.2 Request/Response Examples

POST /api/projects/{id}/calculate

Request:**

Response:**

■ 7. UI/UX Design

7.1 Wireframes

1. Homepage / Landing

2. Input Workflow (Step 1: Building)

3. Input Workflow (Step 2: Materials)

4. Input Workflow (Step 3: Location & Settings)

5. Results Dashboard

6. Heat Loss Breakdown Chart

7. COP vs. Temperature Chart

■■ 8. Technical Stack

■ 9. Project Structure

■ 10. Testing Strategy

10.1 Unit Tests (Backend)

Example Test (pytest):**

from app.core.heat_loss import calculate_heat_loss

def test_heat_loss_walls():

surfaces = [{"u_value": 0.3, "area": 200, "delta_t": 25}]

ventilation = {"air_changes": 0.5, "volume": 250, "delta_t": 25}

result = calculate_heat_loss(surfaces, ventilation)

assert result["total"] == pytest.approx(1500 + 1031.25, rel=0.01) # 1500 (walls) + 1031.25 (vent)

10.3 Frontend Tests

10.4 Validation Strategy

Benchmark Against EnergyPlus/HPDM

• Compare results for **standard test cases** (e.g., reference buildings).

• Target: **±5% accuracy**.

User Testing

• Recruit **5–10 HVAC contractors** for feedback.

• Track **time-to-completion** and **error rates**.

Edge Cases

• Extremely cold climates (-20°C).

• Highly insulated buildings (Passivhaus).

• Large commercial buildings (10,000 m²).

■ 11. Deployment Plan

11.1 Environment Setup

11.2 CI/CD Pipeline (GitHub Actions)

name: Test and Deploy

on:

push:

branches: [main, develop]

pull_request:

```
branches: [main]

jobs:

test-backend:

runs-on: ubuntu-latest

steps:

• uses: actions/checkout@v4

• uses: actions/setup-python@v4

with:

python-version: "3.10"

• run: pip install -r backend/requirements.txt

• run: pytest backend/tests/

test-frontend:

runs-on: ubuntu-latest

steps:

• uses: actions/checkout@v4

• uses: actions/setup-node@v3

with:

node-version: 18

• run: npm install

• run: npm test

working-directory: frontend

deploy-staging:

needs: [test-backend, test-frontend]

if: github.ref == 'refs/heads/develop'

runs-on: ubuntu-latest

steps:

• uses: actions/checkout@v4

• run: vercel --token ${{ secrets.VERCEL_TOKEN }} --env staging

working-directory: frontend

• run: railway up --detach

working-directory: backend
```

env:

RAILWAY_TOKEN: ${{ secrets.RAILWAY_TOKEN }}

deploy-production:

needs: [test-backend, test-frontend]

if: github.ref == 'refs/heads/main'

runs-on: ubuntu-latest

steps:

• uses: actions/checkout@v4

• run: vercel --token ${{ secrets.VERCEL_TOKEN }} --prod

working-directory: frontend

• run: railway up --detach --environment production

working-directory: backend

env:

RAILWAY_TOKEN: ${{ secrets.RAILWAY_TOKEN }}

■ 12. MVP Development Timeline

Total MVP Duration: 24 Weeks (6 Months)**

■ Appendix: Sources & References**

Market Data: Grand View Research, Market Research Future

Existing Tools: GeoT*SOL, Polysun, EnergyPlus, HPDM, TESPy

Standards: BS EN 12831, ASHRAE 90.1

Greenhouse/CEA: Appropedia, PMC/NCBI, ScienceDirect